Rotations and complex numbers

Advanced high school · geometry · Worked solutions · 09

Exercise 1. Find $i(2-3i)$, then its modulus.

Solution. $i(2-3i) = 3+2i$; its modulus is $\sqrt{13}$, the same as that of $2-3i$.

Exercise 2. Find the successive images of $1+i$ under four multiplications by i.

Solution. $-1+i$, $-1-i$, $1-i$, then $1+i$. Four quarter-turns return to the starting point.

Exercise 3. What is the centre of $z' = iz + 2$?

Solution. The centre satisfies $a = ia + 2$, so $a = \frac{2}{1-i} = 1+i$.

Exercise 4. For $z' = a + i(z-a)$, prove $|z'-a| = |z-a|$.

Solution. $z'-a = i(z-a)$, hence $|z'-a| = |i||z-a| = |z-a|$.

Problem · Investigate, justify, conclude. Define $z_0 = 2$ and $z_{n+1} = iz_n + 2$. Find the fixed point a and an explicit formula for z_n. Determine the minimal period and the sum $z_0 + z_1 + z_2 + z_3$. Interpret the four points geometrically.

Solution. $a = 1+i$ and $z_{n+1} - a = i(z_n - a)$, so $z_n = 1 + i + i^n(1-i)$. The values are $2, 2+2i, 2i, 0$, then 2. The first four are distinct, so the minimal period is 4. Their sum is $4+4i = 4a$. The points form a square centred at $(1,1)$, with side 2 and radius $\sqrt{2}$.